

14/12/2019 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
NOV-DEC 2019

Max. Marks: 100

Class: S.Y.BTech

Name of the Course: Relational Database Management System

Course Code: UCEC404

Duration: 3 Hrs

Semester: IV

Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q1 (a)	Explain about DBMS architecture and role its components in detail with diagram . OR Give the differences between 1. File system and Database system 2. Physical and logical Data Independence	10
Q1 (b)	In academic world, A researcher can either be employed as a professor or a lab assistant. There are three kinds of professors: Assistant, associate, and full professors. The following should be stored: 1. For each researcher, his/her name, year of birth, and current position (if any). 2. For each institution, its name, country, and inauguration year. 3. For each institution, the names of its schools (e.g. School of Law, School of Business, School of Computer Science, . . .). A school belongs to exactly one institution. 4. An employment history, including information on all employments (start and end date, position, and what school). 5. Information about co-authorships, i.e., which researchers have co-authored a research paper. The titles of common research papers should also be stored. 6. For each researcher, information on his/her highest degree (BSc, MSc or PhD), including who was the main supervisor, and at what school. 7. For each professor, information on what research projects (title, start date, and end date) he/she is involved in, and the total amount of grant money for which he/she was the main applicant. Draw an ER diagram for the data set described above. Make sure to indicate all cardinality constraints specified above. The ER diagram should not contain redundant entity sets, relationships, or attributes. Also, use relationships whenever appropriate.	10

	OR a. Design a union/category for a motor-vehicle sales company. The company sells motorcycles, passenger cars, vans, and buses. Justify your placement of attributes at each level of the hierarchy. b. Mention the steps to convert ERR model to relational model																						
Q 2 (a)	Discuss the following clauses with examples (i) group by having (ii) alter table (iii) Domain OR Briefly explain security and authorization in SQL.	10																					
Q. 2 (b)	Write the SQL Queries for the given schema Student(Name, Student number, Class, Major) Course(CourseName, CourseNumber, Credit Hours, Department) Section(Sectionidentifier, coursenummer, Semester, Year, Instructor) Grade_report(studentNumber, SectionIdentifier, Grade) a) Change the Major of students from 'IT' to 'COMP' b) Delete the record for the student whose grade is less than 5 c) For each section taught by Prof. Navathe., find the highest grade scored along with the student name. d) Insert a new course, (<'Data science', 'CS1234', 3, 'Comp')	10																					
Q3 (a)	. What is Normalization? Why Normalize is required normalize the below given relation up to highest possible normal form. STUDENT: <table><tr><th>Stu ID</th><th>Stu Name</th><th>City</th><th>Pin code</th><th>Project ID</th><th>Project Name</th><th>Course Content</th></tr><tr><td>S1</td><td>Vijay</td><td>Surat</td><td>326201</td><td>P101</td><td>health programming</td><td>C++, Java, C</td></tr><tr><td>S2</td><td>Kartika</td><td>Pune</td><td>325456</td><td>P102</td><td>social WEB</td><td>HTML, PHP, ASP</td></tr></table>	Stu ID	Stu Name	City	Pin code	Project ID	Project Name	Course Content	S1	Vijay	Surat	326201	P101	health programming	C++, Java, C	S2	Kartika	Pune	325456	P102	social WEB	HTML, PHP, ASP	10
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Q3 (b)	. Explain types Functional dependency with example What is lossless decomposition? Explain the various desirable properties of decomposition.	10																					

Q4 (a)	Explain various types of LOCKs used in Lock base protocol for concurrency control. How does the strict two-phase locking protocol solve problems of concurrency .	10
Q4 (b)	Define Transaction. List out properties of transaction. Explain the different states of transaction	10
Q5 (a)	Consider the three transactions T1, T2, T3 and the schedules S1 and S3 given below. i) State whether schedule is serializable or not. ii) If a schedule is serializable, write down the equivalent serial schedule (s). T1: r1(X); r1(Z); w1(X); T2: r2(Z); r2(Y); w2(Z); w2(Y); T3: r3(X); r3(Y); w3(Y); S1: r1(X); r2(Z); r1(Z); r3(X); r3(Y); w1(X); w3(Y); r2(Y); w2(Z); w2(Y); S2: r1(X); r2(Z); r3(X); r1(Z); r2(Y); r3(Y); w1(X); w2(Z); w3(Y); w2(Y);	10
Q5(b)	Write short notes on any one a. View serializability with example b. Different ways of Transaction recovery	10